An Empirical Investigation of the Relationship between Democracy and Carbon Emissions in Nation-States

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Introduction

Carbon emissions are one of the biggest challenges facing the globe right now. It is claimed to affect climate change and cause global warming. Climate change poses a fundamental threat to the places, species and people's livelihoods (WWF). These unusual patterns of weather change are disturbing the usual balance of nature. The temperature rise is causing the glaciers to melt, posing a threat to the flora and fauna of the region. The area covered by sea ice in the Arctic at the end of summer has shrunk by about 40% since 1979. (NOAA)

Several activities ranging from the burning of fossil fuels to increased consumption are held responsible for the increasing level of carbon dioxide in the atmosphere. According to reports, fossil fuels are the major contributors to global climate change, accounting for a total of 75% of greenhouse gas emissions.

According to the Synthesis Report (SYR) of the IPCC Sixth Assessment Report, Human activities, principally through emissions of greenhouse gases, have unequivocally caused global warming, with the global surface temperature reaching 1.1°C above 1850–1900 in 2011–2020. Unsustainable energy use, land use and land-use change, lifestyles and patterns of consumption and production across regions, between and within countries, and among individuals have all contributed to rising global greenhouse gas emissions.

The role of institutions cannot be ignored while trying to mitigate climate change. Government efforts and policies play a huge role in reducing the carbon footprints of a country. In a democratic country, people have the right to information, freedom to protest and raise their voices for environmental concerns. The government is accountable to its people, so they can form effective climate policies. Several studies suggest that economic freedom helps to mitigate climate change and reduce carbon emissions. This paper attempts to find out the extent to which the political regime affects the level of carbon emission in a country.

Research Questions

The key research questions we aim to answer with this paper are:

I. What is the nature and magnitude of the relationship between democratic governance and carbon emissions, and to what extent does this relationship vary across different countries?

II. What are the mechanisms through which democratic institutions influence carbon emissions in a country, and how do these mechanisms differ in countries with different levels of democratic quality?

III. How do political and institutional factors, such as electoral systems, party systems, and public participation, interact with environmental policy and shape a country's carbon emissions profile?

Literature Review

The study on ‘The impact of economic freedom on the Environment’, concludes that where the reduction of environmental problems is concerned, it must also include the quality of economic freedom. Economic freedom is certainly important in addressing ASEAN's CO2 emissions issues. They propose that the governments of seven ASEAN countries focus more on enhancing economic freedom and restricting countries' authority over economic activity, rather than strengthening control over them. (Setyadharma, S I Nikensari, S Oktavilia and I F S Wahyuningrum).

Increases in economic freedom between 1975 and 1990 are to some extent caused by the level of political freedom. (Jakob de Haan & Jan-Egbert Sturm, 2003). Since democratic regimes provide political freedom, it can be concluded that democratic regimes tend to provide more economic freedom than authoritarian regimes.

In his study on ‘How Political Equality Moderates the Economic Growth-CO_2 Emissions Relationship’, the author finds that Greater political equality mitigates both consumption and production-based emissions. He further suggests that while political equality is a necessary condition, it isn’t sufficient to reduce carbon emissions. (Ryan P. Thomas, 2021) Thus, democracy can be considered an important factor in determining the effectiveness of climate change policies.

In a study on how the rule of law affects carbon emission, the authors found out that autocracies emit more carbon dioxide to reach the same level of per-capita income as vis-a-vis democracy. They took data from 150 countries to conduct their research and quantified the effect of the interaction between autocracy and democracy index on carbon emission. Their analysis further suggests that strengthening law enforcement and access to justice plays an important part in decreasing carbon emissions and it can improve the environment quality. (Apra Sinha, Ashish Sedai, Abhishek Kumar, Rabindra Nepal, 2021). Their paper acts as another strong evidence that a relationship between political regimes and carbon emission exists.

International Institute for Democracy and electoral assistance (IDEA), in the paper 'Democracy and the Challenge of Climate Change) claim that due to their industrialization and high levels of prosperity, the majority of fully democratic nations emit comparable high quantities of greenhouse gases per person. About two-thirds of all historical carbon emissions are attributed to the three major democracies—the European Union, Japan, and the United States. The study mentions that in general, there aren't any significant solutions that democracy presents to deal with the climate crises. Democratic countries are most concerned about the environment within their state boundaries. But still, they have several advantages when it comes to mitigating climate crises. Their policies are more promising than authoritarian regimes. (Danial Lindvall, 2021)

The author also makes policy recommendations on how democracy can be strengthened while fighting against climate change. The author suggests the adoption of climate laws, involvement of the youth of the nation, development of a sustainable market economy, a strong constitutional framework, etc. Acting on climate justice, Knowledge-based decision-making, protection of human rights, and involvement of citizens in the formulation of climate policies are some other important factors that the author suggests should be taken into consideration.

The author also emphasizes the alarming need to do further research in the field because the problem of global warming is only going to get severe in the future. The study suggests that more studies should be conducted on which constitutional set-ups are most efficient in fighting climate crises.

More democratic countries are more receptive to public demands for environmental protection. The authors in the paper 'Democracy and Environmental Quality', argue that the governance of a country is a major determinant of environmental policies and their effectiveness. The importance that policymakers accord to various societal preferences has a bearing on the link between the amount of environmental quality that the public prefers and the level that the State supplies. Therefore, society's choices for environmental quality have an impact on the state's environmental policies. (Y. Hossein Farzin & Craig A. Bond)

All these studies act as strong evidence suggesting that there is a relation between the kind of political regime or governance a country has and the amount of carbon they emit.

Research Methodology

Research design: This study will use a cross-sectional research design, which involves collecting data at a single point in time from a sample of nation-states to investigate the relationship between the democratic index and carbon emissions.

Data collection: Data for this study will be collected from secondary sources, including the World Bank and the Democracy Index by The Economist Intelligence Unit. The sample will consist of a randomly selected group of nation-states from around the world, with a focus on including a diverse range of countries based on geography, income level, and level of democratic quality.

Variables: The dependent variable in this study will be carbon emissions per capita, measured in metric tons obtained by the data repository of the World Bank. The independent variable will be the democratic index, measured using the Democracy Index score provided by The Economist Intelligence Unit.

Analysis: The primary analysis method will be regression analysis, specifically linear regression, to examine the relationship between the independent and dependent variables. The regression equation will be in the form of $Y = \beta 0 + \beta 1X1$, where Y represents carbon emissions per capita and X1 represents the democratic index. The beta coefficients (β) will be estimated using ordinary least squares (OLS) regression, and the significance of the coefficients will be tested using t-tests.

Ethical considerations: Since the data used in this study are publicly available, no ethical considerations are necessary. However, the researcher will ensure the data is properly cited and not misused in any way.

Limitations: One limitation of this study is that it only captures the relationship between democracy and carbon emissions at a single point in time, which may not accurately represent the long-term relationship between these variables. Additionally, the study is limited by the availability and quality of data from secondary sources, which may introduce bias or errors in the analysis.

Diagnostic Tests

Regression Diagnostic	**Test**	**Result**
Heteroscedasticity	Breusch-Pagan test	p-value = 0.98
Autocorrelation	Durbin – Watson test	p-value = 0.77
	Breusch-Godfrey test	p-value = 0.396

The studentized Breusch-Pagan test is used to test for heteroscedasticity in a linear regression model by testing whether the variance of the residuals is constant across different levels of the independent variables. The output of the studentized Breusch-Pagan test shows a test statistic of BP = 0.0002711 with 1 degree of freedom and a p-value of 0.9869. Since the p-value is greater than the usual significance level of 0.05, we fail to reject the null hypothesis of homoscedasticity, which means that there is no evidence of heteroscedasticity in the model. Therefore, we can conclude that there is no significant evidence of heteroscedasticity in your model based on the studentized Breusch-Pagan test.

The Durbin-Watson test is used to test for the presence of autocorrelation in the residuals of a linear regression model. Autocorrelation occurs when there is a correlation between the residuals at different points in time, indicating that the model may be missing an important explanatory variable. The output of the Durbin-Watson test shows a test statistic of DW = 2.1311 and a p-value of 0.7745. Since the p-value is greater than the usual significance level of 0.05, we fail to reject the null hypothesis of no first-order autocorrelation, which means that there is no evidence of autocorrelation in the residuals of the model.

The Breusch-Godfrey test is used to test for the presence of serial correlation in the residuals of a linear regression model. The output of the Breusch-Godfrey test shows a test statistic of LM = 0.72054 and a p-value of 0.396. Since the p-value is greater than the usual significance level of 0.05, we fail to reject the null hypothesis of no first-order serial correlation, which means that there is no evidence of serial correlation in the residuals of the model up to the first order.

Regression Results

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.815240343
R Square	0.664616816
Adjusted R Square	0.662287766
Standard Error	5.157701975
Observations	146

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	7591.105419	7591.105419	285.3596311	5.60019E-36
Residual	144	3830.672111	26.60188966		
Total	145	11421.77753			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	6.672151392	0.442483515	15.07886998	2.00673E-31	5.79754951	7.546753275
Democratic Index	-0.454193072	0.026887117	-16.89259101	5.60019E-36	-0.507337477	-0.401048666

The present study employs econometric analysis to explore the relationship between the dependent variable (Carbon Emissions – metric tons per capita) and the independent variable, namely, the Democratic Index. The results of the analysis reveal that there is a statistically significant and robust linear relationship between the two variables, with a multiple R-value of 0.815, indicating a strong positive correlation. Moreover, the R squared value of 0.664 implies that approximately 66% of the variability in the dependent variable can be explained by the independent variable, reflecting a reasonably high level of explanatory power.

The analysis also shows that the model has a good fit, as demonstrated by the adjusted R squared value of 0.662, which takes into account the number of independent variables in the model. The standard error of 5.1577, which represents the average difference between the predicted and actual values of the dependent variable, is reasonable and indicates that the model has adequate predictive accuracy.

Further analysis, utilizing an ANOVA table, confirms that the regression model is significant at the 5% level of significance, as indicated by the F-statistic of 285.3596. The coefficient of the independent variable, the Democratic Index, is -0.454, which suggests that there is a negative relationship between the independent variable and the dependent variable. Specifically, for every one-unit increase in the independent variable, the dependent variable decreases by 0.454 units. This coefficient is statistically significant at the 5% level of significance, with a 95% confidence interval ranging from -0.507 to -0.401.

Discussion of Results

The Democratic Index by the Economist Intelligence Unit takes into account five key categories: electoral process and pluralism; civil liberties; the functioning of government; political participation; and political culture.

The electoral process and pluralism ensure that governments are held accountable to the public and that the voices of all citizens are heard in the political process. This can lead to policies that reflect the preferences and needs of the broader population, rather than those of a narrow elite. In terms of reducing carbon emissions, this can translate into policies that prioritize investment in renewable energy, regulate emissions from industry, and incentivize energy efficiency.

Civil liberties, including freedom of expression and assembly, are essential for promoting public awareness and engagement on environmental issues. A free press can provide information on environmental problems and promote public debate on how best to address them. This can help to build public support for policies to reduce carbon emissions.

The functioning of government, including the quality of public services and the effectiveness of public institutions, can also play a role in reducing carbon emissions. Effective institutions can facilitate the implementation of policies to reduce emissions, while poor governance can hinder progress in this area.

Political participation, including the ability of citizens to participate in decision-making processes, is essential for promoting the public's engagement in environmental issues. Citizen involvement in policymaking and implementation can increase the legitimacy of environmental policies and help to build support for their implementation.

Finally, political culture, including attitudes towards democracy and the environment, can also play a role in reducing carbon emissions. Countries with a strong culture of environmentalism tend to be more supportive of policies to address climate change, while countries with a weak culture of democracy may be less supportive of policies that prioritize the public good over private interests.

Conclusion: Policy Imperatives

In conclusion, our empirical investigation has demonstrated a significant relationship between democracy and carbon emissions in nation-states. The analysis revealed that democracy, as measured by the Democratic Index by the Economist Intelligence Unit, has a negative effect on carbon emissions per capita. The findings suggest that countries with higher levels of democracy, as measured by the five key categories of the electoral process and pluralism, civil liberties, the functioning of government, political participation, and political culture, tend to have lower carbon emissions per capita.

Our results provide support for the argument that democratic governance can lead to better environmental outcomes by increasing public participation in decision-making, promoting transparency and accountability, and encouraging a more equitable distribution of resources. These findings have important implications for policymakers and stakeholders in the global effort to mitigate climate change. As countries continue to pursue economic growth and development, it is essential that they prioritize democratic governance and environmental sustainability as complementary goals.

Further research could explore the mechanisms underlying the relationship between democracy and carbon emissions, including the role of public participation and civil society in shaping environmental policy. Additionally, future studies could examine how other factors, such as economic development and energy production, interact with democracy to influence carbon emissions. Overall, our study highlights the importance of democratic governance in promoting a more sustainable and equitable future for all.

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Appendix

- R Code to perform diagnostic tests for the model :

```
install.packages("lmtest")


# Load the data from the CSV file
data <- read.csv("RM-project_R.csv")

# Fit the linear regression model
model <- lm(CO2 ~ DEM, data = data)

# Heteroscedasticity test
library(lmtest)
bptest(model)

# Autocorrelation test
library(lmtest)
dwtest(model)
bgtest(model)
```